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EXTENDED ABSTRACT

Realistic decade-long sea-ice simulations from generative models trained for sub-daily forecasting

Tobias S. Finn[1*], Marc Bocquet[1], Pierre Rampal[2], Charlotte Durand[2], Flavia Porro[3], Alban Farchi[4] and Alberto Carrassi[3]

[1]CEREA, ENPC, EDF R&D, Institut Polytechnique de Paris, Marne-la-Vall´ee, 77455, France
[2]Institut des Geosciences de l'Environnement, CNRS, Grenoble, France
[3]Dept. of Physics and Astronomy "Augusto Righi", University of Bologna, Bologna, Italy
[4]ECMWF, European Centre for Medium-Range Weather Forecast, Reading, RG2 9AX, United Kingdom
*Corresponding author. Email: tobias.finn@enpc.fr

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1. Introduction and objectives

The AI revolution has transformed Earth system modelling. Deep neural networks trained on large datasets for short-term forecasts predict the temporal evolution of Earth system components at unprecedented speeds [1, 2], while reaching the accuracy of the best geophysical models. This revolution has also an impact on how we model sea-ice numerically. While numerical models with advanced rheologies [3] can depict the short-term sea-ice dynamics as seen from satellites and drifters, they inflate the computational costs. On the other hand, efficient AI-based models compromise physical realism by staying univariate and by performing only deterministic forecasts [4], leading into smoothing effects and physical inconsistencies as similarly seen for atmospheric models. Amplifying over time, this smoothing and inconsistencies result into drifts, long-term instabilities, and the loss of physical realism. Here, we break this trade-off by employing an efficient generative flow model for sea-ice predictions, and applying it to decade-long simulations.

2. Methods and data

We instantiate a surrogate model to learn to predict the evolution of six different sea-ice state variables (thickness, concentration, sub-grid damage, u-velocity, v-velocity, and snow-on-ice thickness) for a 12-hour lead time given the current state (initial condition) and atmospheric forcings (2-metre temperature, 2-metre specific humidity, 10-metre u-wind velocity and v-wind velocity) over the 12-hour window. The model is a generative flow matching model that, conditioned on the initial conditions and forcings, iteratively maps noise samples to predicted tendencies which, added to the initial conditions, yield the prediction. The neural network that defines the underlying dynamical system of the flow is a transformer with around 30×10^6 parameters. For efficient training and predictions, we use a domain decomposition approach in which we split the global domain into local subdomains on which each iterative step is performed, overlaps thereby facilitate the communication between subdomains. This way, the model has been trained within 2 days on $8\times$ Nvidia A100 GPUs, and we achieve a low memory footprint ($<$ 4

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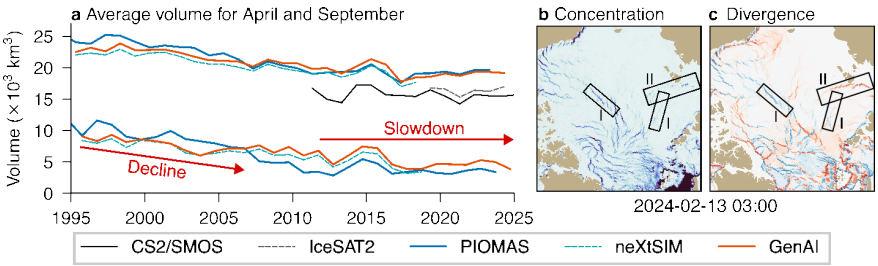

Figure 1. *Trained for 12-hour forecasting, the generative AI sea-ice model is stable for 30 years, shows a similar thermodynamical evolution as numerical models and satellite products, and remains physically consistent even after 30 years, exhibiting (I) opening of leads and (II) ridging among others.*

GB) and a fast inference time (25 simulated years per day), which potentially allows us to perform large ensemble simulations.

The model has been trained on 20 years of coupled sea-ice–ocean simulations from the neXtSIM-NEMO system [5] over the full Arctic at a 12 km resolution, and the atmospheric forcings come from the ERA5 reanalysis product. Once trained, we apply the model autoregressively, where the prediction becomes the next initial condition. We perform a 30-year-long simulation over the full Arctic, starting from 1995-01-01, always updating the atmospheric forcings from the ERA5 reanalysis. The simulation is compared to the geophysical simulation, the PIOMAS reanalysis product, and two satellite products from the combination of CryoSat2 and SMOS and from IceSAT2.

3. Results

The generative AI-based sea-ice model remains stable over the full 30-year-long simulation (Fig. 1a). The temporal evolution of the sea-ice volume, as product of thickness and concentration, is similar to the one observed in the geophysical simulation, the reanalysis, and the satellite product, even though a small bias persists. The seasonal cycle with the maximum in April and minimum in September is captured well, and we can even discover the influence of the sea-ice volume decline due to climate change and the recent slowdown in the decline due to internal variability. Besides the long-term evolution shaped by thermodynamical processes, sea ice is influenced by the dynamics on short time scales. When ice is weaker, it becomes more prone to external stress such as wind, leading to deformation with divergence and convergence, which results in the opening of leads and ridging. In our simulation, we can track these processes even after almost 30-years of simulation (Fig. 1b).

4. Discussion

The model is heavily driven by the atmospheric forcings on long time scales, effectively stabilising the model in addition to the apparent stabilisation due to generative AI. Nonetheless, it remains surprising that the model learns to extract this information from the forcings, as the long-term signal has a vanishing signal-to-noise ratio compared to the short-term signal during the training. Moreover, as the ocean forcings become important on year-long time scales, this additionally suggests that the model implicitly learns to represent ice–ocean interactions that are included in the original data without using ocean forcings. Hence, generative flow models can seemingly learn the full spectrum of information, efficiently generalising from short-term forecasting to long-term simulations.

The thermodynamical evolution can be even learned with deterministic models that smooth out the predictions. However, to obtain the correct dynamics, we need models that retain information at all scales and excel in learning high-dimensional correlations. The combination of generative AI together

with autoregressive forecasting is apparently highly efficient for the generalisation. Consequently, the model remains physically consistent and has implicit internalised physical processes.

Code and model availability

The code to reproduce the results is available on Zenodo [https://zenodo.org/records/17738328, 6] with the trained model checkpoints available on HuggingFace [https://huggingface.co/tobifinn/gensim, 7]. This abstract has a corresponding pre-print [8].

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